Integrating Data Treasures

Knowledge Graphs of DSMZ Digital Diversity

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OUR DATABASES

DSMZ Digital Diversity

The *Digital Diversity Hub* provides integrated information on microbial resources and fosters enhanced access to all other databases and tools.

hub.dsmz.de

BRENDA

The BRENDA Enzyme Database is the world's most comprehensive enzyme information repository and offers a large palette of tools.

brenda-enzymes.de

DSMZ CellDive

DSMZCellDive helps with selecting suitable model cell lines by providing insights into gene expressions and more.

celldive.dsmz.de

PhageDive

PhageDive is a specific database for bacteriophages and archaeal viruses.

phagedive.dsmz.de

TYGS

The Type (Strain) Genome Server (TYGS) is a user-friendly high-throughput web server for genome-based prokaryote taxonomy.

tygs.dsmz.de

StrainInfo

StrainInfo is a service for resolving microbial strain identifiers.

straininfo.dsmz.de

silva

SILVA is a comprehensive, quality-controlled web resource for up-to-date aligned rRNA gene sequences and tools.

arb-silva.de

BacDive

The Bacterial Diversity Database BacDive is the worldwide largest knowledgebase for standardized bacterial phenotypic information.

bacdive.dsmz.de

MediaDive

MediaDive is a manually curated collection of cultivation media and growth conditions.

mediadive.dsmz.de

LPSN .dsmz.de

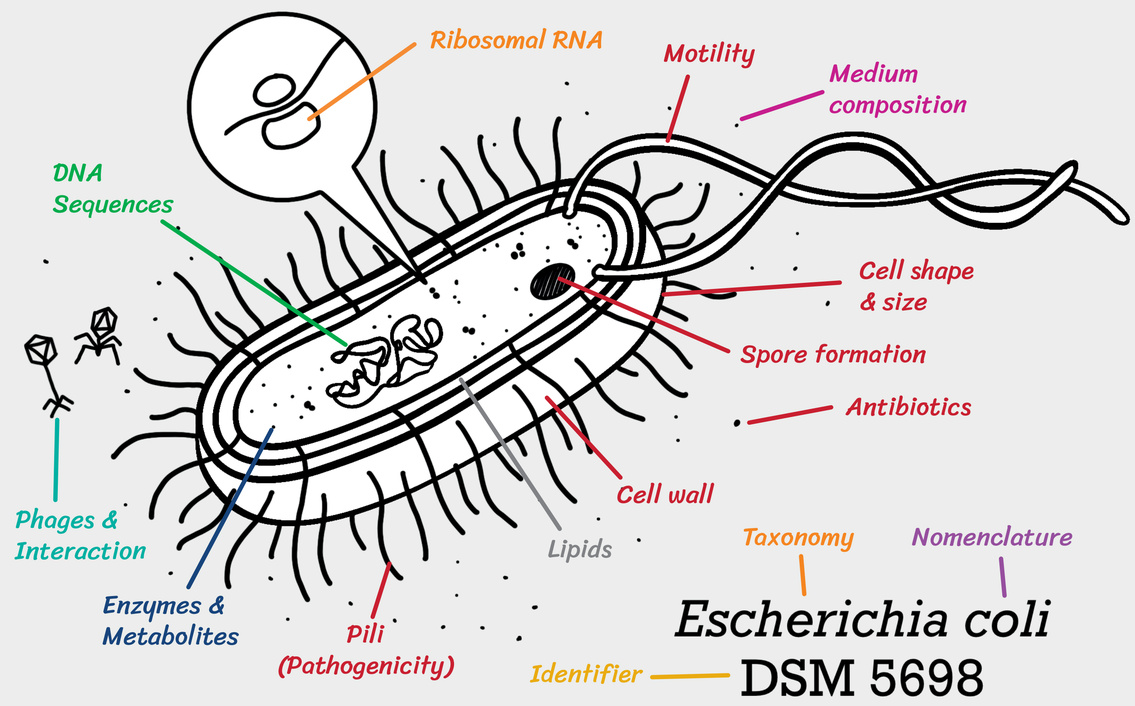
The List of Prokaryotic names with Standing in Nomenclature provides comprehensive information on the nomenclature of prokaryotes and more.

lpsn.dsmz.de

ELIXIR Core Data Resource

Global Core Biodata Resource

THE MICROBIAL DATA TREASURE



Microbes are rich sources of data, spanning genetics, metabolism, and interactions.

DSMZ Digital Diversity offers diverse datasets to create a comprehensive resource for microbiology research. This figure highlights key data types:

- **Genomics & Sequences** — DNA, RNA, and protein sequences
- **Phenotypic Traits** — Cell shape, motility, sporulation, and antibiotic resistance
- **Metabolism & Biochemistry** — Enzymes, metabolites, and media composition
- **Taxonomy & Nomenclature** — Systematics and classification
- **Microbial Interactions** — Environmental niche and phage-host relationships

By linking these data sources, we enable deeper insights and new discoveries in microbiology. Learn more about DSMZ Digital Diversity on:

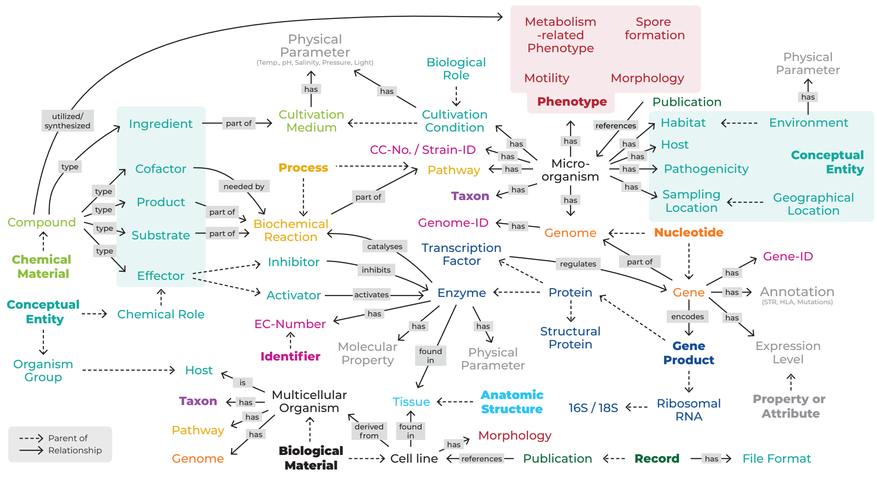
hub.dsmz.de

BacDive · SILVA · StrainInfo · TYGS · FATB · PhageDive · BRENDA · LPSN · MediaDive

THE INTEGRATION

Digital Diversity Knowledge Graphs

Knowledge graphs will be developed based on existing and new ontologies and the different data fields we cover.



Knowledge Graph on Bacterial Trait Data

sparql.dsmz.de/bacdive

The BacDive Knowedge Graph contains trait data on almost 100,000 prokaryotic strains, including taxonomy, morphology, physiology, isolation source, genome sequences, etc.

Knowledge Graph on Cultivation Media

sparql.dsmz.de/mediadive

The MediaDive Knowledge Graph contains more than 3,300 cultivation media and growth conditions for more than 46,000 strains from all domains of life.

NEXT STEPS ON THIS JOURNEY

Ontology Integration

We strive for semantic consistency and reuse of well-established vocabularies and provide new ontologies where needed.

Data Compliance

We currently update all of our database models to comply with existing standards and terminologies.

"Plug & Play" Knowledge Graphs

We are building comprehensive knowledge graphs that will be interoperable by using shared vocabularies.

SPARQL & Usability

We want to make querying and exploring life science knowledge graphs more intuitive.

THE TEAM

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DZIF Deutsches Zentrum für Infektionsforschung

NFDI 4 Biodiversity